A COMPETENCY-ASSESSMENT HANDBOOK

Miller's Pyramid for Rasashastra & Bhaishajya Kalpana

Module-by-module competency mapping, assessment design and usage guide for AYPG-AB-RB — Applied Basics of Rasashastra and Bhaishajya Kalpana
MD/MS Ayurveda, Semester II

Curriculum	NCISM, Board of Ayurveda — applicable from the 2024–25 batch
Coverage	8 modules · 42 units · 171 session learning objectives · 47 practical training and 44 experiential learning activities
Framework	Miller (1990) four tiers, with the Cruess *Is* tier (2016), mapped onto NCISM Table 3, Column 3F

16	480	400	300
CREDITS	NOTIONAL HOURS	FORMATIVE MARKS	SUMMATIVE MARKS

Created by Dr Milind Chatrabhuji using Claude Opus. · Prepared July 2026 · For departmental use in curriculum delivery, assessment design and paper setting.

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PART A

Foundations

Five short chapters. The first explains what Miller's pyramid is and, just as importantly, what it is not. The second shows that it is already inside your curriculum, in a column most teachers have never had pointed out. The third gives the three rules of constructive alignment, the fourth sets out which instrument can certify which tier, and the fifth explains how to read a unit in Part B. Read straight through once; after that, use the modules.

1. **Miller's pyramid** — what it is, and what it is not
2. **The pyramid is already inside your curriculum** — reading Column 3F
3. **Constructive alignment** — the three rules
4. **The assessment toolkit**, tier by tier
5. **How to use a module in Part B**

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PART A · CHAPTER 1

Miller's pyramid — what it is, and what it is not

In 1990 George Miller, a medical educator, published a single diagram that has shaped assessment ever since. He was answering a question every examiner faces: *what does this test actually prove?*

His answer was a pyramid of four tiers, each demanding evidence more authentic than the one below.

- **Knows** — the student can recall it.
- **Knows how** — the student can reason with it, apply it to a case on paper.
- **Shows how** — the student can do it when someone is watching, in a set-up arranged for the purpose.
- **Does** — the student does it in real practice, unprompted, when nobody is checking.

The tiers are cumulative but not interchangeable. A student who reaches *Does* necessarily passed through the others; a student who has only been tested at *Knows* has been told nothing about the tiers above. That asymmetry is the whole point. **Evidence from a lower tier never certifies a higher one.**

WHAT THE PYRAMID IS NOT

It is not a taxonomy of thinking. It says nothing about whether a task is easy or hard, or whether it calls for recall or judgement. A *Knows* objective can be intellectually demanding and a *Does* objective can be routine. Bloom's taxonomy answers "what kind of thinking is this?"; Miller answers "how authentic must the evidence be?" The two are independent axes, and this curriculum carries both — Bloom in one column, Miller in another. Each has its own companion volume.

It is also not a ranking of worth. A curriculum that stops at *Shows how* for a dangerous procedure is not being lazy; it is making a judgement about supervision and safety. Reading the ceilings as a scoreboard is the commonest misuse of the pyramid.

> **The Cruess addition.** In 2016 Richard and Sylvia Cruess proposed a fifth tier above *Does*: **Is** — professional identity. Not what the student does but who they have become. It cannot be examined in a hall, and it is not coded anywhere in your curriculum. It is worth naming because some of what you certify at *Does* — a safety briefing recorded honestly, a result reported when it was inconvenient — is really evidence of *Is*. Where a unit in Part B says so, that is what it means.

WHY IT MATTERS HERE

This semester sits high on the pyramid: 78 of its 171 objectives are at *Shows how* or *Does* — 46%. Nearly half of what this course certifies cannot be certified on paper at all, and the credit framework says so openly by spending two thirds of every credit outside the lecture room.

PART A · CHAPTER 2

The pyramid is already inside your curriculum

Miller is not something this book adds to your curriculum. It has been there since publication, in a column most teachers have never had pointed out.

COLUMN 3F

Open Table 3. The column headed **Level** carries one of four values against every objective: *Know*, *Knows how*, *Shows how* or *Does*. Those are Miller's four tiers, and they are the curriculum's statement of how authentic the evidence must be before that objective can be called met.

Across the 171 objectives of this semester the distribution is:

TIER	OBJECTIVES	SHARE
Knows	14	8.2%
Knows how	79	46.2%
Shows how	35	20.5%
Does	43	25.1%

Total 171 objectives.

Read that against the hours. Every credit is 5 hours of lecture, 10 of practical training and 13 of experiential learning. The upper two tiers and the lower two are almost evenly split, and so are the hours meant to build them.

> **A word about the other column headed "Domain".** Beside the Level column sits another carrying codes like CK, CC, CAP, PSY-MEC and AFT-VAL. Those are Bloom's taxonomy and the psychomotor and affective domains — a different question entirely, and the subject of the companion volume. In *this* book, "level" and "tier" always mean Miller.

THE CEILING

Every module in this curriculum has a **ceiling**: the highest tier any of its objectives reaches. That single fact governs assessment more than anything else on the page. It tells you what evidence you must gather, and it tells you what you may not ask for. Part B prints it for every module.

PART A · CHAPTER 3

Constructive alignment — the three rules

Constructive alignment is the principle that what you teach, what you ask students to do, and what you accept as proof must all sit at the same tier. Stated as rules, it comes to three.

RULE 1 — THE TIER DECIDES THE INSTRUMENT

Not the topic, not the hours, not the marks. Once you know an objective sits at *Shows how*, the set of instruments that can certify it is fixed, and a written paper is not among them. Chapter 4 gives the toolkit tier by tier.

RULE 2 — NEVER ASSESS ABOVE THE CEILING

If the curriculum certifies a module to *Knows how*, do not build an OSPE station for it. You would be examining something the students were never taught to that standard, and the marks would measure prior exposure rather than your teaching. This rule protects students.

RULE 3 — NEVER CERTIFY A HIGHER TIER WITH LOWER-TIER EVIDENCE

This is the rule that gets broken, and it is broken in one direction. A *Does* objective assessed by viva has not been assessed. A *Shows how* objective assessed by a long answer has not been assessed. The student may pass, the record may look complete, and nothing has been established.

> **The asymmetry is the thing to remember.** You may always gather evidence from a tier *above* the one required — a student observed doing the thing has plainly also shown how. You may never accept evidence from a tier *below*. Every alignment error worth catching is a violation of that one sentence.

WORKING THE THREE RULES BACKWARDS

In practice the fastest way to plan a session is to start at the end. Ask what evidence you will hold at the finish — a script, a checklist, a log entry. That names the tier. The tier names the instrument, and the instrument names the teaching that must precede it. A session planned forwards from the syllabus tends to arrive at the assessment it can most easily set, which is almost always a written one.

PART A · CHAPTER 4

The assessment toolkit, tier by tier

Each tier admits a particular set of instruments. Using one from the wrong tier is the single commonest fault in assessment design, and it is invisible unless you look for it, because the paperwork is complete either way.

TIER	THE QUESTION IT ANSWERS	TEACHING THAT BUILDS IT	INSTRUMENTS THAT CERTIFY IT	EVIDENCE YOU KEEP
Knows	**Can the student recall it?**	Lecture, reading, recitation, retrieval practice.	MCQ, short answer, oral quiz, spotting.	Written script, quiz sheet.
Knows how	**Can the student reason with it?**	Worked cases, discussion, problem-based work; the student must reason aloud and be corrected.	Structured long answer, case-based questions, structured viva.	Reasoned written answer, viva score sheet.
Shows how	**Can the student do it when watched?**	Demonstration, then supervised practice with your help withdrawn a little at a time.	OSPE or OSCE station, observed performance against a checklist.	Checklist score, annotated artefact, station record.
Does	**Does the student do it unprompted, in real practice?**	Real work with real consequences, supervised at a distance rather than at the elbow.	Log book, record book, portfolio, observation over time, presentation with peer feedback.	Dated log entry, countersigned record, portfolio piece.

THE TRAP AT EACH TIER

- **Knows** — Spending long-answer marks here. A long answer cannot separate a student who knows the list from one who knows it better.
- **Knows how** — Setting a question answerable from memory. If the stem can be met by recall, the objective has silently dropped to Knows.
- **Shows how** — Certifying by viva. A fluent student who has never handled the material will pass a viva comfortably.
- **Does** — Certifying with a written test. Nothing written can show that a student does something unprompted when nobody is watching.

> **The instruments this curriculum names.** This curriculum sets a modular assessment in every module and requires a different method each time, with a record of the pattern used. It also runs a university examination at the end of the semester: one theory paper of 100 marks and a practical examination of 200.

ON THE LOG BOOK AND RECORD BOOK

These are the only instruments in routine use that can carry *Does*-tier evidence, and they are usually treated as paperwork. A log filled in at the end of term is a piece of writing. The same log filled in as the work happens, dated and countersigned, is the only proof you will have that a student did something unprompted. Decide at the start of each module what artefact each *Does* objective will leave behind, and the record becomes evidence rather than an administrative task.

PART A · CHAPTER 5

How to use a module in Part B

Part B has eight modules, each carrying its units inside it. Each begins on its own page and is laid out the same way, so once you have read one you can use any of them.

WHAT IS ON THE PAGE

1. **The heading and counters** — hours, objectives, marks, and the **ceiling**: the highest tier this module reaches. Read the ceiling first; it governs everything else.
2. **The session map** — the curriculum's own objectives with the tier each is rated to, so you can see at a glance where the weight sits.
3. **The pyramid for this module** — four rows, one per tier. Each names what a student must be able to do at that tier here, then how to build it, how to assess it, and what evidence you keep. Tiers the curriculum does not use are shown greyed, because knowing what is *not* certified matters as much as knowing what is.
4. **The ceiling statement** — a single line naming the highest tier and warning against assessing above it or certifying it with the wrong instrument.
5. **Two panels** — one for the teacher, one to give to students.

THREE WAYS TO USE IT

Planning a session. Start at the ceiling row of the pyramid and work down: that row tells you what the students must end up able to do, and the rows beneath tell you what has to be in place first.

Setting a paper. A written paper can honestly reach *Knows* and *Knows how* and no further. Use the session map to see which objectives those are; anything above must be certified elsewhere, and the module says where.

Auditing what you already do. Take your existing assessment for a module and ask which tier it can certify. If it is a written test and the ceiling is *Shows how* or *Does*, the gap is your finding.

> **What this book does not do.** It does not rewrite the curriculum and does not tell you what to teach — every objective in Part B appears in the curriculum's own words. What it adds is the Miller layer: what standard of evidence each objective demands, and which instruments can supply it.

PART B

The modules

Eight modules, one per module, each carrying its five units inside it. Every objective appears in the curriculum's own words, gathered under the tier its Level column gives it, with the kind of hour beside it. The pyramid block on each module names what every tier asks for in that module, how to build it, how to assess it and what evidence to keep. Chapter 5 explains the layout.

1. **Modules** — hours, tier counts, the pyramid, and the ceiling
2. **Unit strips** — each unit's objectives, grouped by tier

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MODULE 1

Pharmaceutics and Pharmaceutical Technology

2 credits | 60 notional hours | 5 units | **24 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	13	5	6
not used	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	10	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	8	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	6	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
6 objectives

Identify applications suiting various regulatory frameworks; Explore, identify, and collect various classical and modified pharmaceutical dosage forms; Identify an Appropriate Excipient According to Solid and Liquid Dosage Form; and 3 more.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
5 objectives

Demonstrate regulatory frameworks (FDA, etc.); Demonstrate pharmaceutical product development process; Formulate Solid Dosage Forms selecting appropriate excipients; and 2 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
13 objectives

Define pharmaceutics and describe its scope and historical development; Discuss Pharmaceutical industry overview, market size (national and international) and regulatory framework; Enlist and evaluate challenges and solutions in the pharmaceutical product development process; and 10 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This module does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Introduction to Pharmaceutics

6 obj · 12 h · ceiling Does

Definition and scope of pharmaceutics
Historical development of pharmaceutics
Pharmaceutical industry overview
Regulatory frameworks (FDA, etc.)
Pharmaceutical product development process

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Define pharmaceutics and describe its scope and historical development.	Lecture	1	Knows how
Discuss Pharmaceutical industry overview, market size (national and international) and regulatory framework.	Lecture	1	Knows how
Enlist and evaluate challenges and solutions in the pharmaceutical product development process.	Experiential Learning	3	Knows how
SHOWS HOW — 2			
Demonstrate regulatory frameworks (FDA, etc.).	Practical Training	2	Shows how
Demonstrate pharmaceutical product development process.	Practical Training	2	Shows how
DOES — 1			
Identify applications suiting various regulatory frameworks.	Experiential Learning	3	Does

Unit 2 · Pharmaceutical Dosage Forms

4 obj · 8 h · ceiling Does

Classification of dosage forms
Solid dosage forms (tablets, capsules, etc.)
Liquid dosage forms (solutions, suspensions, etc.)
Semi-solid dosage forms (ointments, creams, etc.)
Parenteral dosage forms (injections, infusions, etc.)

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Classify various pharmaceutical dosage forms.	Lecture	1	Knows how
Identify and categorise classical Ayurvedic dosage forms.	Practical Training	2	Knows how
Identify and categorise modified Ayurvedic dosage forms.	Practical Training	2	Knows how
DOES — 1			
Explore, identify, and collect various classical and modified pharmaceutical dosage forms.	Experiential Learning	3	Does

Unit 3 · Pharmaceutical Formulation

5 obj · 16 h · ceiling Does

Principles of pharmaceutical formulation
Excipients and their functions
Solid dosage forms
Liquid and semi-solid dosage forms
Parenteral dosage forms

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Describe the principles of pharmaceutical formulations, excipients, and their functions.	Lecture	1	Knows how
Discuss formulation of parenteral dosage forms.	Lecture	1	Knows how
SHOWS HOW — 2			
Formulate Solid Dosage Forms selecting appropriate excipients.	Practical Training	4	Shows how
Formulate liquid Dosage Forms selecting appropriate excipients.	Practical Training	4	Shows how

DOES — 1			
Identify an Appropriate Excipient According to Solid and Liquid Dosage Form.	Experiential Learning	6	Does

Unit 4 · Pharmaceutical Manufacturing and Quality Control

5 obj · 10 h · ceiling Does

Good Manufacturing Practices (GMP& cGMP)
Quality control and quality assurance
Packaging and labelling
Portability and storage

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss Good Manufacturing Practices (GMP& cGMP).	Lecture	1	Knows how
Discuss pharmaceutical stability and shelf life.	Lecture	1	Knows how
SHOWS HOW — 1			
Demonstrate, design and evaluate pharmaceutical packing and labelling.	Practical Training	2	Shows how
DOES — 2			
Evaluate GMP audit and compliance.	Practical Training	2	Does
Design and develop Pharmaceutical Packaging systems and labelling.	Experiential Learning	4	Does

Unit 5 · Biopharmaceutics

4 obj · 10 h · ceiling Does

Biopharmaceutics and pharmacokinetics
Bioavailability and bioequivalence
Pharmaceutical dissolution and absorption
Pharmaceutical product development and optimization

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Describe biopharmaceutics and pharmacokinetics.	Lecture	1	Knows how
Describe bioavailability and bioequivalence.	Lecture	1	Knows how
Discuss pharmaceutical product development and optimization.	Lecture	1	Knows how
DOES — 1			
Design and Optimize Pharmaceutical Formulation for Enhanced Bioavailability.	Experiential Learning	7	Does

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **11 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 2

Databases and Information Systems for Metal, Mineral, and Plant Research

2 credits | 60 notional hours | 5 units | **22 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
1	9	2	10
certified here	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	10	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	7	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	5	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
10 objectives

Demonstrate various searching techniques of databases; Demonstrate Modelling, Querying, and Analytics of Data; Analyze USGS Mineral Database; and 7 more.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
2 objectives

Design and Implement a Simple Database for a Plant Nursery; Analyse case studies of Database Applications in Metal, Mineral, or Plant Research.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
9 objectives

Describe overview of databases and information systems, types of databases (relational object-oriented, etc.), and database management systems (DBMS); Describe data retrieval and querying languages (SQL, etc.) and an overview of metal mineral, and plant databases; Describe overview of metal and mineral databases, USGS (U.S. Geological Survey); and 6 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
1 objective

Identify Database-Related Challenges in Pharmaceutical Development.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Introduction to Databases and Information Systems

5 obj · 12 h · ceiling Does

- Overview of databases and information systems
- Types of databases (relational, object-oriented, etc.)
- Database management systems (DBMS)
- Data retrieval and querying languages (SQL, etc.)

Introduction to metal, mineral, and plant databases

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Describe overview of databases and information systems, types of databases (relational object-oriented, etc.), and database management systems (DBMS).	Lecture	1	Knows how
Describe data retrieval and querying languages (SQL, etc.) and an overview of metal mineral, and plant databases.	Lecture	1	Knows how
SHOWS HOW — 1			
Design and Implement a Simple Database for a Plant Nursery.	Experiential Learning	6	Shows how
DOES — 2			
Demonstrate various searching techniques of databases.	Practical Training	2	Does
Demonstrate Modelling, Querying, and Analytics of Data.	Practical Training	2	Does

Unit 2 · Metal and Mineral Databases

6 obj · 16 h · ceiling Does

Overview of metal and mineral databases
USGS Mineral Database
Mineral Database
Webmineral Mineral Database
Introduction to Metal Databases

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Describe overview of metal and mineral databases, USGS (U.S. Geological Survey).	Lecture	1	Knows how
Discuss Mineral and metal Databases, including their applications and significance.	Lecture	1	Knows how
DOES — 4			
Analyze USGS Mineral Database.	Practical Training	2	Does
Analysis and integration of Web mineral Database.	Practical Training	3	Does
Search, explore and analyze Metal Database and Applications.	Practical Training	3	Does
Analyse Web mineral Database and Metal Database.	Experiential Learning	6	Does

Unit 3 · Plant Databases

4 obj · 12 h · ceiling Does

Overview of plant databases
The Plant List (TPL)
International Plant Names Index (IPNI)
Royal Botanic Gardens, Kew (RBG Kew) databases
Introduction to plant genetic databases (e.g., PlantGDB)

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Illustrate overview of plant databases, The Plant List (TPL), and International Plant Names Index (IPNI).	Lecture	1	Knows how
Describe Royal Botanic Gardens, Kew (RBG Kew) databases and introduction to plant genetic databases (e.g., PlantGDB).	Lecture	1	Knows how
DOES — 2			
Demonstrate techniques to explore Plant Databases.	Practical Training	4	Does
Explore and analyse plant database.	Experiential Learning	6	Does

Unit 4 · Database application

4 obj · 12 h · ceiling Does

Database applications in research and industry

Using databases for decision-making and policy development

Future directions: emerging trends and technologies in database development

Group project: developing a database application for metal, mineral, or plant research

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss database applications in research and industry, and use databases for decision-making and policy development.	Lecture	1	Knows how
Analyze future directions in database development and applications, including emerging trends and technologies, and group project guidelines.	Lecture	1	Knows how
SHOWS HOW — 1			
Analyse case studies of Database Applications in Metal, Mineral, or Plant Research.	Practical Training	4	Shows how
DOES — 1			
Apply and justify case studies of Database Applications in Metal, Mineral, or Plant Research.	Experiential Learning	6	Does

Unit 5 · Challenges and limitation of Databases

3 obj · 4 h · ceiling Does

Challenges in Pharmaceutical Formulation Development

Limitations of In Vitro and In Vivo Testing

Regulatory Challenges in Pharmaceutical Product Approval

Scalability and Manufacturing Challenges

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Identify Database-Related Challenges in Pharmaceutical Development.	Lecture	1	Knows
KNOWS HOW — 1			
Discuss data Quality, Integration, Security, Scalability, and Regulatory Compliance.	Lecture	1	Knows how
DOES — 1			
Perform a role play related to the challenges and limitations of databases.	Experiential Learning	2	Does

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **12 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 3

Essential Chemistry :- Unlocking the Rasashastra Techniques

2 credits | 60 notional hours | 5 units | **21 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
5	7	4	5
certified here	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	10	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	5	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	6	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
5 objectives

Apply the knowledge Periodic Table for Chemical Properties of Main Group and Transition Metals; Demonstrate extraction of different herbal formulations using various extraction techniques and solvents; Identify phytoconstituents in Herbs and Herbal Formulations: A; and 2 more.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
4 objectives

Analyse chemical bonding types and principles and isomerism and their significance in better understanding of different Rasashastra processes such as Shodhana, and Marana; Demonstrate different types of extraction methods including solvent selection; Demonstrate qualitative identification of phytoconstituents in single herbs and herbal formulations; and 1 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
7 objectives

Demonstrate the essential fundamental principles of crystallography and organometallic principles with special reference to Rasashastra procedures; Discuss the interaction of herbs and animal products with metals; Explore and identify different functional groups and discuss their influence on molecular interactions and therapeutic efficacy; and 4 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
5 objectives

Describe the role of basic chemistry principles and their application in Rasashastra; Describe the principles of crystallography and an overview of organometallic and bioinorganic chemistry; Describe the structure of organic molecules; and 2 more.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Atomic, Molecular Structure, and Periodic Table Elements

3 obj · 12 h · ceiling Does

Importance of knowledge of chemistry for understanding various processing methods of Rasashastra and Oushadha nirmana.

Atomic structure, Periodic table, Chemical bonding, Isomerism -Application of these principles in Shodhana, Marana, Amruteekarana and Chaturvidha Rasayana

Periodic Table, Main group elements, Transition metals: understanding of chemical properties of non-metals, metals, variable oxidation states, thermal stability,

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Describe the role of basic chemistry principles and their application in Rasashastra.	Lecture	1	Knows
SHOWS HOW — 1			
Analyse chemical bonding types and principles and isomerism and their significance in better understanding of different Rasashastra processes such as Shodhana, and Marana.	Practical Training	5	Shows how
DOES — 1			
Apply the knowledge Periodic Table for Chemical Properties of Main Group and Transition Metals.	Experiential Learning	6	Does

Unit 2 · Specialized Chemistry Fields-An Outline

2 obj · 6 h · ceiling Knows how

Crystallography- detoxification, transformation of compound

Organometallic chemistry- Chaturvidha Rasayana

Bioinorganic chemistry- Amruteekarana (Metal-Carbon bond-stable compound formation)

Application in Shodhana, Marana, Amruteekarana, Chaturvidha Rasayana

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Describe the principles of crystallography and an overview of organometallic and bioinorganic chemistry.	Lecture	1	Knows
KNOWS HOW — 1			
Demonstrate the essential fundamental principles of crystallography and organometallic principles with special reference to Rasashastra procedures.	Practical Training	5	Knows how

Unit 3 · Organic Chemistry

3 obj · 8 h · ceiling Knows how

Structure of organic molecules: understanding of Herbal components,

Interaction of herbs and animal products with metals for detoxification, compound formation, and compound stability.

Functional groups: Reactivity and properties of organic molecules influencing compound action

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Describe the structure of organic molecules.	Lecture	1	Knows
KNOWS HOW — 2			
Discuss the interaction of herbs and animal products with metals.	Lecture	1	Knows how
Explore and identify different functional groups and discuss their influence on molecular interactions and therapeutic efficacy.	Experiential Learning	6	Knows how

Unit 4 · Phytochemistry

7 obj · 16 h · ceiling Does

Definition and significance of phytochemistry
Overview of plant-based bioactive compounds
Classes of Phytochemicals
Extraction methods
Phytochemical screening

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Describe the significance of phytochemistry.	Lecture	1	Knows how
Describe the overview of plant-based bioactive components.	Lecture	1	Knows how
Discuss different types of extraction methods including solvent selection.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate different types of extraction methods including solvent selection.	Practical Training	4	Shows how
Demonstrate qualitative identification of phytoconstituents in single herbs and herbal formulations.	Practical Training	3	Shows how
DOES — 2			
Demonstrate extraction of different herbal formulations using various extraction techniques and solvents.	Experiential Learning	3	Does
Identify phytoconstituents in Herbs and Herbal Formulations: A.	Experiential Learning	3	Does

Unit 5 · Metallurgy

6 obj · 14 h · ceiling Does

Introduction to metallurgy and general physicochemical methods of extraction of metals from ores
Common ores used in Rasashastra and their Satvapatana from metallurgical point of view

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 2			
Define metallurgy and describe its scope.	Lecture	1	Knows
Describe general physicochemical methods of extraction of metals from ores.	Lecture	1	Knows
KNOWS HOW — 1			
Discuss common ores used in Rasashastra and their Satvapatana with metallurgical point of view.	Lecture	1	Knows how
SHOWS HOW — 1			
Demonstrate different types of metals and their properties.	Practical Training	3	Shows how
DOES — 2			
Differentiate various types of metals based on their properties.	Experiential Learning	4	Does
Differentiate metals using basic metal identification techniques.	Experiential Learning	4	Does

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **9 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 4

Concept, understanding, application, SOPs and chemical changes of Shodhana and Marana procedures

2 credits | 60 notional hours | 5 units | **15 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	11	3	1
not used	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	10	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	2	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	3	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
1 objective

Demonstrate Shodhana procedure of Rasadravyas using various methods.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
3 objectives

Perform Shodhana of Rasadravyas with various methods; Demonstrate Marana procedure of Rasadravyas with different methods; Perform Marana procedure of Rasadravyas with different methods.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
11 objectives

Discuss Shodhana and Marana procedures; Discuss the importance of purification and incineration in Ayurvedic pharmaceutical manufacturing; Analyse principles of Shodhana and Marana procedures with their application in Ayurvedic pharmaceutical manufacturing; and 8 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This module does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Shodhana and Marana Procedures

3 obj · 3 h · ceiling Knows how

Overview of Shodhana and Marana procedures

Importance of purification and incineration in Ayurvedic pharmaceutical manufacturing

Principles of Shodhana and Marana procedures

Applications of Shodhan and Maran procedures in Ayurvedic pharmaceutical manufacturing

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Discuss Shodhana and Marana procedures.	Lecture	1	Knows how
Discuss the importance of purification and incineration in Ayurvedic pharmaceutical manufacturing.	Lecture	1	Knows how
Analyse principles of Shodhana and Marana procedures with their application in Ayurvedic pharmaceutical manufacturing.	Lecture	1	Knows how

Unit 2 · SOP of Shodhana Processes

4 obj · 22 h · ceiling Does

Types of Shodhana procedures (physical, chemical, biological)

Methods of Shodhana (filtration, sedimentation, distillation, etc.)

SOPs for Shodhana procedures

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Describe types of Shodhana procedures (physical, chemical, biological).	Lecture	1	Knows how
Discuss methods of Shodhana (filtration, sedimentation, distillation, etc.) and its SOPs.	Lecture	1	Knows how
SHOWS HOW — 1			
Perform Shodhana of Rasadravyas with various methods.	Experiential Learning	10	Shows how
DOES — 1			
Demonstrate Shodhana procedure of Rasadravyas using various methods.	Practical Training	10	Does

Unit 3 · SOP of Marana process

4 obj · 22 h · ceiling Shows how

Types of Marana procedures (incineration, calcination, etc.)

Methods of Marana (oxidation, thermal decomposition, etc.)

SOPs for Marana procedures

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Describe types of Marana procedures (incineration, calcination, etc.).	Lecture	1	Knows how
Discuss methods of Marana (oxidation, thermal decomposition, etc.) and its SOPs.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate Marana procedure of Rasadravyas with different methods.	Practical Training	10	Shows how
Perform Marana procedure of Rasadravyas with different methods.	Experiential Learning	10	Shows how

Unit 4 · Chemical Changes involved in Shodhana and Marana processes

1 obj · 1 h · ceiling Knows how

Chemical changes involved in Shodhan procedures

Chemical Changes involved in Marana procedures

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Discuss chemical changes involved in Shodhana and Marana procedures.	Lecture	1	Knows how

Unit 5 · Shodhana and Marana- Case Studies, troubleshooting and emerging trends

3 obj · 8 h · ceiling Knows how

Case studies of Shodhana and Marana procedures in Ayurvedic pharmaceutical manufacturing

Troubleshooting and optimization of Shodhana and Marana procedures

Future directions and emerging trends in Shodhana and Marana procedures

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Describe Troubleshooting and optimization of Shodhana and Marana procedures.	Lecture	1	Knows how
Discuss future directions and emerging trends in Shodhana and Marana procedures.	Lecture	1	Knows how
Analyse and write Shodhana and Marana procedures in Ayurvedic pharmaceutical manufacturing through case studies.	Experiential Learning	6	Knows how

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **4 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 5

Principles, specification, SOPs of handling of classical and advanced instruments

3 credits | 90 notional hours | 7 units | **29 objectives** | 75 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
1	18	6	4
certified here	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	15 h	15	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	30 h	7	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	39 h	7	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 6 hours carry the modular assessment, worth 75 marks.

THE PYRAMID FOR THIS MODULE

Does
4 objectives

Demonstrate selected Puta to give the knowledge of Standard Operative Procedure; Develop Standard Operative Procedures (SOPs)for given Puta; Demonstrate chromatography techniques to separate, and identify components and interpretation of results; and 1 more.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
6 objectives

Demonstrate the handling of Musha following the Standard Operating Procedure (SOP); Develop Standard Operative Procedure for handling Musha; Perform chromatography to separate and identify components and interpret the results; and 3 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
18 objectives

Discuss the principles and specifications of Puta; Analyse modern aspects of Puta; Discuss the principles and specifications of Musha; and 15 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
1 objective

Describe the specifications of spectroscopy instruments.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Puta and its modern aspects

4 obj · 14 h · ceiling Does

Introduction, principles, and specifications of Puta

SOPs for handling Puta

Modern aspects of Puta (automation, computerization, etc.)

Troubleshooting and maintenance of Puta

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss the principles and specifications of Puta.	Lecture	1	Knows how
Analyse modern aspects of Puta.	Lecture	1	Knows how
DOES — 2			
Demonstrate selected Puta to give the knowledge of Standard Operative Procedure.	Practical Training	4	Does
Develop Standard Operative Procedures (SOPs)for given Puta.	Experiential Learning	8	Does

Unit 2 · Musha and its modern aspects

4 obj · 14 h · ceiling Shows how

Introduction, principles, and specifications of Puta.

SOPs for handling Musha

Modern aspects of Musha (automation, computerization, etc.)

Troubleshooting and maintenance of Musha

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss the principles and specifications of Musha.	Lecture	1	Knows how
Analyse modern aspects of Musha with special reference to crucibles.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate the handling of Musha following the Standard Operating Procedure (SOP).	Practical Training	4	Shows how
Develop Standard Operative Procedure for handling Musha.	Experiential Learning	8	Shows how

Unit 3 · Chromatography

5 obj · 17 h · ceiling Does

Scope and applications of chromatography in herbal drug standardization (identification of active compounds, adulterant detection, phytochemical analysis, and quality as

Principle and types of chromatography, including TLC, HPTLC, HPLC, LC-MS, and GC-MS.

Instrumentation of chromatography, including equipment and tools used in different types of chromatography.

Data analysis in chromatography, including interpretation of chromatograms and spectra.

Chromatography's Limitations, challenges, and prospects in pharmaceutical and phytochemical research

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Discuss the scope and applications of chromatography in herbal drug standardization.	Lecture	1	Knows how
Describe the instrumentation of chromatography, including equipment and tools used in different types of chromatography.	Lecture	1	Knows how
Discuss chromatography's Limitations, challenges, and prospects in pharmaceutical and phytochemical research.	Lecture	1	Knows how
SHOWS HOW — 1			
Perform chromatography to separate and identify components and interpret the results.	Experiential Learning	8	Shows how

DOES — 1			
Demonstrate chromatography techniques to separate, and identify components and interpretation of results.	Practical Training	6	Does

Unit 4 · Spectroscopy

4 obj · 9 h · ceiling Knows how

Introduction, principles, and types (IR, UV, NMR, AAS, ICP, etc.) of spectroscopy.

Specifications of spectroscopy instruments

SOPs for handling spectroscopy instruments

Troubleshooting and maintenance of spectroscopy instruments

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Describe the specifications of spectroscopy instruments.	Lecture	1	Knows
KNOWS HOW — 3			
Discuss spectroscopy's principles and types (IR, UV, NMR, AAS, ICP, etc.).	Lecture	1	Knows how
Demonstration of spectroscopic instruments through Audiovisual aids.	Practical Training	3	Knows how
Explore and analyze published articles on spectroscopic instruments.	Experiential Learning	4	Knows how

Unit 5 · Microscopy

4 obj · 12 h · ceiling Shows how

Introduction, principles, and types of microscopy (optical, petrological, electron, etc.)

Specifications of microscopy instruments

SOPs for handling microscopy instruments

Use of micrometer in microscopy

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Describe the principles, types, and specifications of microscopy instruments.	Lecture	1	Knows how
Describe the Specifications of microscopy instruments.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate handling of petrological and trinocular microscopes.	Practical Training	6	Shows how
Perform hands-on practice on the trinocular microscope.	Experiential Learning	4	Shows how

Unit 6 · Calibration techniques

3 obj · 7 h · ceiling Does

Introduction, principles, and types of calibration (instrument calibration, method calibration, etc.)

SOPs for calibration

Troubleshooting and maintenance of calibration instruments

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Discuss principles and types of calibration.	Lecture	1	Knows how
SHOWS HOW — 1			
Demonstrate calibration of various instruments present in the department.	Practical Training	3	Shows how
DOES — 1			
Analyse articles related to challenges in the calibration of equipment.	Experiential Learning	3	Does

Unit 7 · Overview of characterization instruments

5 obj · 11 h · ceiling Knows how

Overview of advanced characterization instruments and their significance in the standardization of Ayurvedic medicines.

Principles and applications of working mechanism and Applications of instrumental methods: PSA, XRD, XRF, SEM-E-Dax, TEM, TGA, DTA, VSM analysis

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 5			
Discuss overview of Advanced Characterization Instruments and their significance in the standardization of Ayurvedic medicines.	Lecture	1	Knows how
Discuss principles and Applications of Spectroscopic and X-ray techniques (PSA, XRD and XRF techniques).	Lecture	1	Knows how
Describe the Principles and Applications of Microscopic and Thermal Techniques(SEM-EDAX, TEM, TGA, DTA, and VSM).	Lecture	1	Knows how
Demonstrate characterization instruments with Audiovisual aids.	Practical Training	4	Knows how
Analyse published articles on advanced instruments used for characterization.	Experiential Learning	4	Knows how

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **10 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 75 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 6

Animal house and Animal handling technique

1 credits | 30 notional hours | 5 units | **13 objectives** | 25 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
1	5	2	5
certified here	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	5 h	5	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	10 h	5	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	13 h	3	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 2 hours carry the modular assessment, worth 25 marks.

THE PYRAMID FOR THIS MODULE

Does
5 objectives

Present structure and functioning of an animal house; Develop critical thinking and research skills by analyzing online resources and studies on experimental animal behavior; Demonstrate Species-Specific Handling Techniques for Laboratory Animals; and 2 more.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
2 objectives

Design and Set Up an Animal House; Observe and. Interpret Animal Body Language through demonstration.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
5 objectives

Describe the principles of Animal House Management with its design and layout; Identify behavioral patterns and body language in animals; Describe Common diseases in laboratory animals and their management; and 2 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
1 objective

Discuss the handling techniques for various laboratory animal species, ensuring handler safety.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Introduction to animal house management

3 obj · 7 h · ceiling Does

Overview of animal house management
Record-keeping and documentation
Animal house design and layout

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Describe the principles of Animal House Management with its design and layout.	Lecture	1	Knows how
SHOWS HOW — 1			
Design and Set Up an Animal House.	Practical Training	2	Shows how
DOES — 1			
Present structure and functioning of an animal house.	Experiential Learning	4	Does

Unit 2 · Study of animal behaviour

3 obj · 8 h · ceiling Does

Introduction to Animal Behaviour
Types of Animal behaviour.

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Identify behavioral patterns and body language in animals.	Lecture	1	Knows how
SHOWS HOW — 1			
Observe and. Interpret Animal Body Language through demonstration.	Practical Training	2	Shows how
DOES — 1			
Develop critical thinking and research skills by analyzing online resources and studies on experimental animal behavior.	Experiential Learning	5	Does

Unit 3 · Species-specific handling techniques

2 obj · 3 h · ceiling Does

Non-human primates
Other specialized species (e.g., birds, reptiles)
Aquatic animals

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Discuss the handling techniques for various laboratory animal species, ensuring handler safety.	Lecture	1	Knows
DOES — 1			
Demonstrate Species-Specific Handling Techniques for Laboratory Animals.	Practical Training	2	Does

Unit 4 · Animal health and disease prevention

3 obj · 7 h · ceiling Does

Common diseases in laboratory animals
Parasite control and management
Vaccination and immunization programs

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Describe Common diseases in laboratory animals and their management.	Lecture	1	Knows how
Prepare and present a parasite Control in Laboratory Animals.	Experiential Learning	4	Knows how

DOES — 1			
Demonstrate microscopic Examination of Faecal Samples for Parasites.	Practical Training	2	Does

Unit 5 · Animal welfare and ethics

2 obj · 3 h · ceiling Does

Animal welfare principles and ethics

Humane endpoints and euthanasia methods

Case studies and group discussions on animal welfare and ethics

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Discuss Animal Welfare Principles, Ethics, Humane Endpoints, Euthanasia Methods, and Case Studies in Animal Research.	Lecture	1	Knows how
DOES — 1			
Evaluate Animal Welfare and Ethics in a Research Setting.	Practical Training	2	Does

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **7 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 25 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 7

Classical and Advanced quality control techniques for Ayurveda Formulations

2 credits | 60 notional hours | 5 units | **26 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
1	9	6	10
certified here	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	10	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	8	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	8	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
10 objectives

Practice documentation and record-keeping processes in the quality control section; Assess Bhasma Siddhi Lakshana (Quality tests to confirm the perfect preparation of Bhasma); Demonstrate quality control parameters for solid dosage forms like tablets, pills, and Varti; and 7 more.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
6 objectives

Demonstrate documentation and record-keeping processes in the quality control section; Demonstrate assessment of Bhasma Siddhi Lakshana (Quality tests to confirm the perfect preparation of Bhasma); Demonstrate Quality control parameters for powder dosage forms like Churna (including Kwatha Churna and Lepa Churna), Kshara, Lavana, Satva; and 3 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
9 objectives

Describe the importance of Quality Control and Quality Assurance in Ayurveda; Discuss Quality Control Parameters, Regulatory Requirements, and Documentation; Discuss Mana and their application in quality control and Bhasma/ Pishti pariksha; and 6 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
1 objective

Discuss quality control parameters for powder dosage forms like Churna (including Kwatha Churna and Lepa Churna), Kshara, Lavana, Satva.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Quality Control and Quality Assurance

4 obj · 6 h · ceiling Does

Importance of quality control in Ayurveda
Quality Assurance of Ayurveda Products
Quality control parameters (identity, purity, etc.)
Regulatory requirements for quality control
Documentation and record-keeping

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Describe the importance of Quality Control and Quality Assurance in Ayurveda.	Lecture	1	Knows how
Discuss Quality Control Parameters, Regulatory Requirements, and Documentation.	Lecture	1	Knows how
SHOWS HOW — 1			
Demonstrate documentation and record-keeping processes in the quality control section.	Practical Training	2	Shows how
DOES — 1			
Practice documentation and record-keeping processes in the quality control section.	Experiential Learning	2	Does

Unit 2 · Classical aspects of the quality control parameter for Ayurvedic formulations

4 obj · 6 h · ceiling Does

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss Mana and their application in quality control and Bhasma/Pishti pariksha.	Lecture	1	Knows how
Discuss Siddhi Lakshana of Avaleha, Snehapaka, Sandhana and Guggulu Paka.	Lecture	1	Knows how
SHOWS HOW — 1			
Demonstrate assessment of Bhasma Siddhi Lakshana (Quality tests to confirm the perfect preparation of Bhasma).	Practical Training	2	Shows how
DOES — 1			
Assess Bhasma Siddhi Lakshana (Quality tests to confirm the perfect preparation of Bhasma).	Experiential Learning	2	Does

Unit 3 · Quality Control parameter for solid dosage forms

9 obj · 26 h · ceiling Does

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Discuss quality control parameters for powder dosage forms like Churna (including Kwatha Churna and Lepa Churna), Kshara, Lavana, Satva.	Lecture	1	Knows
KNOWS HOW — 2			
Discuss Quality control parameters for solid dosage forms like tablets, pills, and Varti.	Lecture	1	Knows how
Discuss Quality control parameters for Khalveeya, Parpati and Kupipakwa Rasayana Lauha/ Mandura.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate Quality control parameters for powder dosage forms like Churna (including Kwatha Churna and Lepa Churna), Kshara, Lavana, Satva.	Practical Training	3	Shows how
Demonstrate quality control parameters for Khalveeya, Parpati and Kupipakwa Rasayana Lauha/Mandura.	Practical Training	4	Shows how

DOES — 4			
Demonstrate quality control parameters for solid dosage forms like tablets, pills, and Varti.	Practical Training	3	Does
Perform Quality control procedures for powder dosage forms like Churna (including Kwatha Churna and Lepa Churna), Kshara, Lavana, Satva.	Experiential Learning	4	Does
Perform quality control procedures for solid dosage forms.	Experiential Learning	4	Does
Perform quality control procedures for Rasoushadhi.	Experiential Learning	5	Does

Unit 4 · Quality Control parameter for semi-solid dosage forms

3 obj · 6 h · ceiling Does

Rasakriya/Avaleha/Paka
Anjana

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Describe Quality control parameters for semisolid dosage forms like Rasakriya/Avaleha/Paka/Anjana as per Pharmacopeial standards.	Lecture	1	Knows how
SHOWS HOW — 1			
Demonstration of Quality control parameters for semisolid dosage forms like Rasakriya/Avaleha/Paka/Anjana as per Pharmacopeial standards.	Practical Training	2	Shows how
DOES — 1			
Perform quality control tests for semisolid dosage forms.	Experiential Learning	3	Does

Unit 5 · Quality Control parameter for liquid and gaseous dosage forms

6 obj · 12 h · ceiling Does

Kashaya
Arka
Sharkara
Asava-arishta
Netra Bindu/Karna Bindu
Aerosols/sprays

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss Quality control parameters for liquid dosage forms like Kashaya, Arka Sharakara, Asava-arishta.	Lecture	1	Knows how
Discuss Quality control parameters for Netra Bindu/Karna Bindu and aerosols/sprays.	Lecture	1	Knows how
SHOWS HOW — 1			
Demonstrate quality control parameters for liquid dosage forms like Kashaya, Arka Sharakara, Asava-arishta.	Practical Training	2	Shows how
DOES — 3			
Demonstrate quality control parameters for Netra Bindu/Karna Bindu and aerosols/sprays.	Practical Training	2	Does
Perform quality control tests for liquid dosage forms.	Experiential Learning	3	Does
Perform quality control tests for Netra Bindu/Karna Bindu and aerosols/sprays.	Experiential Learning	3	Does

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **16 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 8

Nano-science, nano-technology and its application in Ayurveda

2 credits | 60 notional hours | 5 units | **21 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
5	7	7	2
certified here	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	10	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	5	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	6	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
2 objectives

Analyze the impact of Bhavana by measuring the particle size of samples; Conduct a market Survey of Nano-Based Ayurvedic Formulations.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
7 objectives

Demonstrate the measurement of micro-particles using relevant equipment; Demonstrate particle size of Bhasma prepared with different numbers of Puta; Demonstrate particle size of various Kupipakwa Rasayana samples; and 4 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
7 objectives

Describe the basic principles and applications of nanotechnology; Discuss an overview of Ayurvedic Nano formulations and their limitations; Identify the potential of nanotechnology in Ayurvedic formulations; and 4 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
5 objectives

Define nanotechnology and describe the history of nanotechnology; Review articles regarding the green synthesis of metallic nanoparticles; Discuss regulatory frameworks for nanotechnology-based Ayurvedic formulations; and 2 more.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Introduction to Nanotechnology

3 obj · 6 h · ceiling Shows how

Definition and history of nanotechnology

Understanding Nanoscale

Basic principles and applications of nanotechnology

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Define nanotechnology and describe the history of nanotechnology.	Lecture	1	Knows
KNOWS HOW — 1			
Describe the basic principles and applications of nanotechnology.	Lecture	1	Knows how
SHOWS HOW — 1			
Demonstrate the measurement of micro-particles using relevant equipment.	Practical Training	4	Shows how

Unit 2 · Ayurvedic Formulations and Nanotechnology

10 obj · 34 h · ceiling Does

Overview of Ayurvedic formulations and their limitations

Potential of nanotechnology in Ayurvedic formulations

Characterization micro/nanoparticles in Bhasma

Characterization micro/nanoparticles in Kupipakwa Rasayana and Pottali

Characterization of micro-nano sized particles generated during Bhavana

Synthesize silver nanoparticles (AgNPs) using the chemical precipitation method and grasp the fundamental principles underlying nanoparticle synthesis

Market Survey of Nano-Based Herbal Formulations

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Review articles regarding the green synthesis of metallic nanoparticles.	Experiential Learning	2	Knows
KNOWS HOW — 2			
Discuss an overview of Ayurvedic Nano formulations and their limitations.	Lecture	1	Knows how
Identify the potential of nanotechnology in Ayurvedic formulations.	Lecture	1	Knows how
SHOWS HOW — 5			
Demonstrate particle size of Bhasma prepared with different numbers of Puta.	Practical Training	4	Shows how
Demonstrate particle size of various Kupipakwa Rasayana samples.	Practical Training	4	Shows how
Synthesize silver nanoparticles (AgNPs) using the chemical precipitation method.	Practical Training	4	Shows how
Measure particle size of Bhasma prepared with different numbers of Puta.	Experiential Learning	4	Shows how
Measure particle size of various Kupipakwa Rasayana samples.	Experiential Learning	4	Shows how
DOES — 2			
Analyze the impact of Bhavana by measuring the particle size of samples.	Experiential Learning	4	Does
Conduct a market Survey of Nano-Based Ayurvedic Formulations.	Experiential Learning	6	Does

Unit 3 · Nanocarriers for Ayurvedic Formulations

3 obj · 6 h · ceiling Shows how

Types of nanocarriers (liposomes, nanoparticles, etc.)
Advantages and limitations of nanocarriers in Ayurvedic formulations
Conversion of Taila into Oil in water nano emulsion and its characterization

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss types of nanocarriers (liposomes, nanoparticles, etc.).	Lecture	1	Knows how
Discuss the advantages and limitations of nanocarriers in Ayurvedic formulations.	Lecture	1	Knows how
SHOWS HOW — 1			
Demonstrate converting Taila into an oil-in-water nanoemulsion and its characterization.	Practical Training	4	Shows how

Unit 4 · Regulatory Aspects and Safety Concerns

2 obj · 2 h · ceiling Knows

Regulatory frameworks for nanotechnology-based Ayurvedic formulations
Safety concerns and toxicity issues related to nanotechnology-based Ayurvedic formulations

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 2			
Discuss regulatory frameworks for nanotechnology-based Ayurvedic formulations.	Lecture	1	Knows
Discuss safety concerns and toxicity issues related to nanotechnology-based Ayurvedic formulations.	Lecture	1	Knows

Unit 5 · Recent Advances and Future Directions

3 obj · 8 h · ceiling Knows how

Recent advances in nanotechnology-based Ayurvedic formulations
Future directions and potential applications
Exploring the Future of Nanomedicine with Ayurvedic Formulations

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Discuss recent advances in nanotechnology-based Ayurvedic formulations.	Lecture	1	Knows
KNOWS HOW — 2			
Discuss future directions and potential applications of nanotechnology in Ayurvedic formulations.	Lecture	1	Knows how
Integrate nanomedicine with Ayurvedic formulations and their prospects.	Experiential Learning	6	Knows how

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **9 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PART C

Reference

Four appendices, generated from the same data as Part B so the two cannot disagree. C1 decodes the tiers and names the instruments that certify each. C2 gives the ceiling of every module in one view — the fastest way to check that an assessment matches what it is meant to certify. C3 lists the instruments the curriculum itself names, sorted by the tier each can honestly reach. C4 records the sources and what was and was not altered.

1. **C1** The four tiers, and what certifies each
2. **C2** Ceiling of every module
3. **C3** The instruments this curriculum names, by tier
4. **C4** Sources, and a note on how this book was made

↑ Contents

PART C · APPENDIX C1

The four tiers, and what certifies each

Miller's four tiers with the count of objectives at each, the instruments that can certify them, and the evidence you keep. Nothing below a tier can certify it.

TIER	OBJECTIVES	INSTRUMENTS	EVIDENCE
Knows	14	MCQ, short answer, oral quiz, spotting.	Written script, quiz sheet.
Knows how	79	Structured long answer, case-based questions, structured viva.	Reasoned written answer, viva score sheet.
Shows how	35	OSPE or OSCE station, observed performance against a checklist.	Checklist score, annotated artefact, station record.
Does	43	Log book, record book, portfolio, observation over time, presentation with peer feedback.	Dated log entry, countersigned record, portfolio piece.

TIER	OBJECTIVES	SHARE
Knows	14	8.2%
Knows how	79	46.2%
Shows how	35	20.5%
Does	43	25.1%

Total 171 objectives.

PART C · APPENDIX C2

Ceiling of every module

The highest tier each module reaches. Read across: the ceiling names the evidence you must gather, and the tier above it names what you may not ask for.

M	U	UNIT	OBJ	CEILING
MODULE 1 — PHARMACEUTICS AND PHARMACEUTICAL TECHNOLOGY — 24 OBJECTIVES, CEILING DOES				
1	1	Introduction to Pharmaceutics	6	Does
1	2	Pharmaceutical Dosage Forms	4	Does
1	3	Pharmaceutical Formulation	5	Does
1	4	Pharmaceutical Manufacturing and Quality Control	5	Does
1	5	Biopharmaceutics	4	Does
MODULE 2 — DATABASES AND INFORMATION SYSTEMS FOR METAL, MINERAL, AND PLANT RESEARCH — 22 OBJECTIVES, CEILING DOES				
2	1	Introduction to Databases and Information Systems	5	Does
2	2	Metal and Mineral Databases	6	Does
2	3	Plant Databases	4	Does
2	4	Database application	4	Does
2	5	Challenges and limitation of Databases	3	Does
MODULE 3 — ESSENTIAL CHEMISTRY :- UNLOCKING THE RASASHASTRA TECHNIQUES — 21 OBJECTIVES, CEILING DOES				
3	1	Atomic, Molecular Structure, and Periodic Table Elements	3	Does
3	2	Specialized Chemistry Fields-An Outline	2	Knows how
3	3	Organic Chemistry	3	Knows how
3	4	Phytochemistry	7	Does
3	5	Metallurgy	6	Does
MODULE 4 — CONCEPT, UNDERSTANDING, APPLICATION, SOPS AND CHEMICAL CHANGES OF SHODHANA AND MARANA PROCEDURES — 15 OBJECTIVES, CEILING DOES				
4	1	Shodhana and Marana Procedures	3	Knows how
4	2	SOP of Shodhana Processes	4	Does
4	3	SOP of Marana process	4	Shows how
4	4	Chemical Changes involved in Shodhana and Marana processes	1	Knows how
4	5	Shodhana and Marana- Case Studies, troubleshooting and emerging trends	3	Knows how
MODULE 5 — PRINCIPLES, SPECIFICATION, SOPS OF HANDLING OF CLASSICAL AND ADVANCED INSTRUMENTS — 29 OBJECTIVES, CEILING DOES				
5	1	Puta and its modern aspects	4	Does
5	2	Musha and its modern aspects	4	Shows how
5	3	Chromatography	5	Does
5	4	Spectroscopy	4	Knows how
5	5	Microscopy	4	Shows how
5	6	Calibration techniques	3	Does
5	7	Overview of characterization instruments	5	Knows how

M	U	UNIT	OBJ	CEILING
MODULE 6 — ANIMAL HOUSE AND ANIMAL HANDLING TECHNIQUE — 13 OBJECTIVES, CEILING DOES				
6	1	Introduction to animal house management	3	Does
6	2	Study of animal behaviour	3	Does
6	3	Species-specific handling techniques	2	Does
6	4	Animal health and disease prevention	3	Does
6	5	Animal welfare and ethics	2	Does
MODULE 7 — CLASSICAL AND ADVANCED QUALITY CONTROL TECHNIQUES FOR AYURVEDA FORMULATIONS — 26 OBJECTIVES, CEILING DOES				
7	1	Quality Control and Quality Assurance	4	Does
7	2	Classical aspects of the quality control parameter for Ayurvedic formulations	4	Does
7	3	Quality Control parameter for solid dosage forms	9	Does
7	4	Quality Control parameter for semi-solid dosage forms	3	Does
7	5	Quality Control parameter for liquid and gaseous dosage forms	6	Does
MODULE 8 — NANO-SCIENCE, NANO-TECHNOLOGY AND ITS APPLICATION IN AYURVEDA — 21 OBJECTIVES, CEILING DOES				
8	1	Introduction to Nanotechnology	3	Shows how
8	2	Ayurvedic Formulations and Nanotechnology	10	Does
8	3	Nanocarriers for Ayurvedic Formulations	3	Shows how
8	4	Regulatory Aspects and Safety Concerns	2	Knows
8	5	Recent Advances and Future Directions	3	Knows how

Module ceilings are shown in the coloured band; unit ceilings in the rows beneath. A unit may sit below its module's ceiling, and most do.

PART C · APPENDIX C3

The instruments this curriculum names, by tier

Every assessment abbreviation that appears in the curriculum, sorted by the highest tier it can honestly certify.

HIGHEST TIER IT REACHES	INSTRUMENTS THE CURRICULUM NAMES	WHAT IT CAN AND CANNOT SHOW
Knows	MCQ, Spotting, T-OBT	Recall only. Cannot show reasoning or performance.
Knows how	LAQ, SAQ, Structured viva, ABQ, Case study	Reasoning on paper or aloud. Cannot show that the student can perform.
Shows how	OSPE, Practical examination, Checklist	Performance when watched. The lowest instrument that settles a doing objective.
Does	Log book, Record book, Portfolio, Presentation	Independent practice over time. The only evidence that reaches the top tier.

Counts are how many objectives name each instrument. An instrument may always be used below its tier; it may never be used above it.

PART C · APPENDIX C4

Sources, and a note on how this book was made

The primary sources for the framework, the curriculum this book is built on, and a note on what was and was not altered.

THE FRAMEWORK

- Miller, G. E., 'The assessment of clinical skills / competence / performance', *Academic Medicine*, 65(9), 1990. The original four-tier paper.
- Cruess, R. L., Cruess, S. R. and Steinert, Y., 'Amending Miller's Pyramid to Include Professional Identity Formation', *Academic Medicine*, 91(2), 2016. The proposed fifth tier, *Is*.
- Biggs, J., 'Enhancing teaching through constructive alignment', *Higher Education*, 32, 1996. The source of the alignment principle in Chapter 3.

A NOTE ON HOW THIS BOOK WAS MADE

Every objective in Part B is reproduced in the curriculum's own words. Plain-English spelling errors have been corrected and sentence punctuation lost to the source document's line wrapping has been restored. No Sanskrit or technical term has been altered, and no subject content has been added, removed or reworded.

National Commission for Indian System of Medicine, Board of Ayurveda. *Curriculum for MD/MS Ayurveda, Semester II — Applied Basics of Rasashastra and Bhaishajya Kalpana (AYPG-AB-RB).* New Delhi: NCISM. Tables 2, 3, 4, 5 and 6 are the source of every objective, tier, hour and mark in this book. All 171 objectives, 91 activities and 480 notional hours reconcile against the curriculum's own credit framework.

The counts and tables in Part C are generated from the same parsed data as Part B, so the two cannot disagree.